

	learning rate	# epoch	batch size	prefix length
Prefix:				
E2E	8e-05	5	10	5
WebNLG	5e-05	5	5	5
DART	5e-05	10	5	10
XSUM	5e-05	30	14	100
Adapter:				
E2E (3%)	5e-05	5	5	-
E2E (0.1%)	8e-05	10	5	-
WebNLG (3%)	5e-05	5	5	-
WebNLG (0.1%)	5e-05	10	5	-
DART (3%)	5e-05	5	5	-
DART (0.1%)	8e-05	5	5	-
Fine-tune:				
E2E	5e-05	5	10	-
WebNLG	1e-05	10	6	-
DART	1e-05	10	6	-
FT-top2:				
E2E	5e-05	5	10	-
WebNLG	5e-05	10	9	-
DART	5e-05	5	5	-

Table 5: Hyperparameter settings for our method and baseline methods.

table contents). In particular, prefix-tuning tends to undergenerate whereas fine-tuning tends to generate untruthfully. For seen categories, both perform fairly well in terms of coverage and truthfulness.

A Supplementary Material

A.1 Hyperparameters

In Table 5, we report the hyperparameters used to train the models documented in the experiment section.

A.2 Additional Results for Low-data Settings

Figure 6 supplements the low-data performance curves in Figure 3 by plotting the relationship between training size and generation metrics for both prefix-tuning and fine-tuning.

A.3 Additional Results for the Initialization Experiment

Figure 7 supplements Figure 3 by plotting additional metrics for our initialization technique §7.4. It validates that random initialization (from a uniform (0,1) distribution) significantly underperforms initializing with real words; Additionally, initializing with task-relevant words (e.g., “summarization” and “table-to-text”) attains slightly better generation scores than initializing with task-irrelevant words (e.g., “elephant” and “banana”).

A.4 Qualitative Examples for Extrapolation

Table 6 contains qualitative examples from both seen and unseen categories in WebNLG. We find that for unseen categories, both prefix-tuning and fine-tuning tend to undergenerate (generated output do not cover full table contents) or generate untruthfully (generated output is inconsistent with

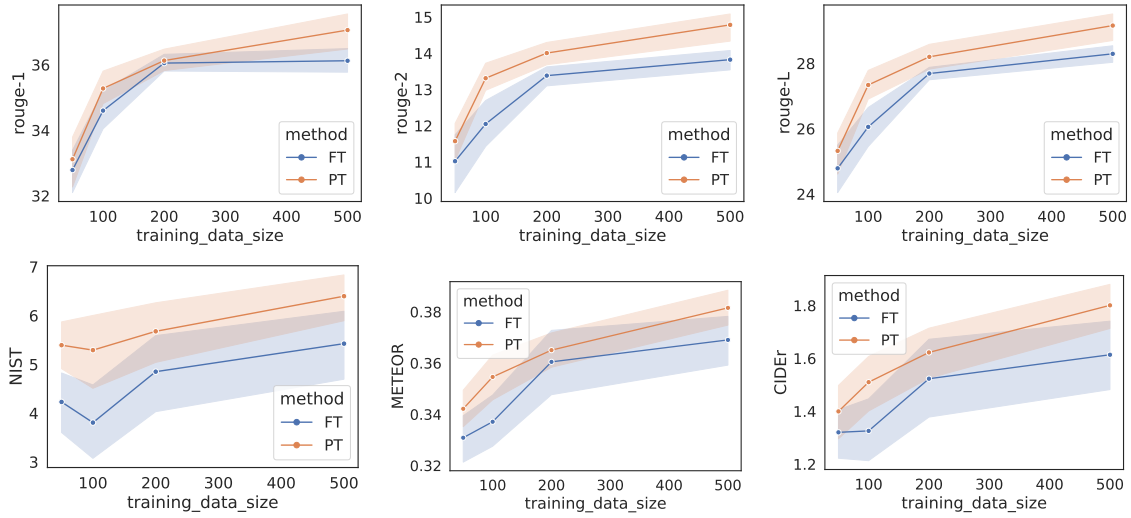


Figure 6: Prefix-tuning (orange) outperforms fine-tuning (blue) in low-data regimes in addition to requiring many fewer parameters. The top three plots correspond to summarization, measured by ROUGE-1, ROUGE-2, and ROUGE-L. The bottom three plots correspond to table-to-text, measured by NIST, METEOR, and CIDEr. The x-axis is the training size and the y-axis is the evaluation metric (higher is better).

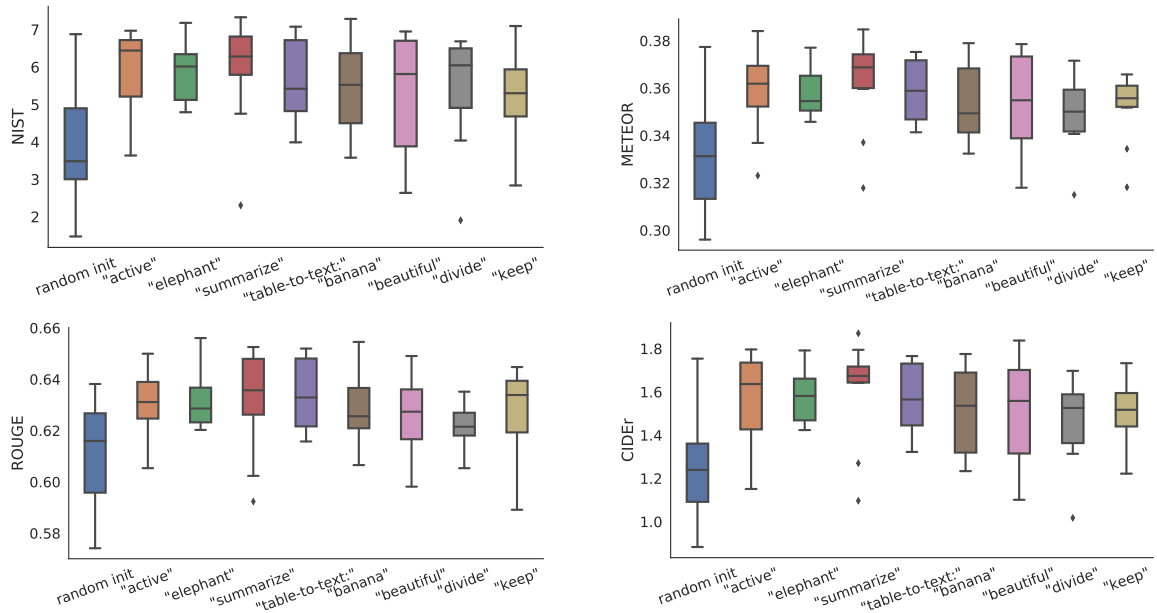


Figure 7: Initializing the prefix with activations of real words significantly outperforms random initialization, in a low-data setting with 100 training data.

Source [Unseen, Athlete]	(Al Kharaitiyat SC, club, Alaa Abdul-Zahra), (Al Khor, ground, Al Kharaitiyat SC), (Shabab Al-Ordon Club, club, Alaa Abdul-Zahra) (Amar Osim, manager, Al Kharaitiyat SC)
Prefix-tuning	Al Kharaitiyat SC are managed by Amar Osim and play at their ground at Al Khor. Al Kharaitiyat SC are also the club for which Alaa Abdul-Zahra is a player.
Fine-tuning	Alaa Abdul-Zahra plays for Al-Kharaitiyat SC and Shabab Al-Ordon Club. He also plays for Al-Khor and manages Al-Kharaitiyat SC.
Reference	Alaa Abdul Zahra plays for Al Kharaitiyat SC which is located at Al Khor and managed by Amar Osim. The Shabab Al-Ordon club is associated with Alaa Abdul-Zahra.
Source [Unseen, Transportation]	(Genoa, location, Costa Crociere), (AIDA Cruises, operator, AIDAstella), (Costa Crociere, owner, AIDAstella)
Prefix-tuning	AID Astella is operated by Aida Cruises and is owned by the Costa Rican tourist resort of Genoa.
Fine-tuning	AID Astella, operated by AIDA-Cruises, is located in Genoa and is owned by the Costa Rican government.
Reference	Costa Crociere is the owner of the AIDAstella and are based in Genoa. The operator of AIDAstella is AIDA Cruises.
Source [Unseen, Politician]	(Euro, currency, Netherlands), (Stellendam, birthPlace, Ab Klink), (Netherlands, nationality, Ab Klink)
Prefix-tuning	Ab Klink was born in Stellendam and is a national of the Netherlands where the currency is the Euro.
Fine-tuning	Ab Klink is a national of the Netherlands where the currency is the Euro. He was born in Stellendam.
Reference	Ab Klink was born in Stellendam in the Netherlands, where the national currency is the euro.
Source [Unseen, Politician]	(Robert E. Lee, commander, Battle of Salem Church), (American Civil War, isPartOfMilitaryConflict, Battle of Salem Church), (Battle of Salem Church, battles, Aaron S. Daggett)
Prefix-tuning	Robert E. Lee was the commander of the Battle of Salem Church which was part of the military conflict in the American Civil war.
Fine-tuning	The Battle of Salem Church is part of the American Civil War and was commanded by Robert E. Lee.
Reference	Robert E Lee was a commander in the Battle of Salem Church, which was one of the military conflicts in the American Civil War. Aaron S Daggett fought in the same battle.
Source [Unseen, Artist]	(Christian alternative rock, musicSubgenre, Alternative rock), (Alternative rock, genre, Andrew White (musician))
Prefix-tuning	Andrew White is a Christian alternative rock musician.
Fine-tuning	Andrew White, a Christian alternative rocker, performs.
Reference	The musician Andrew White's genre is alternative rock, the genre which has the sub genre Christian alternative rock.
Source [Unseen, Artist]	(Hip hop music, genre, Allen Forrest), (solo singer, background, Allen Forrest)
Prefix-tuning	Allen Forrest is a solo singer.
Fine-tuning	Born in
Reference	Allen Forrest is a solo singer whose genre is Hip Hop music.
Source [Seen, ComicsCharacter]	(Americans, nationality, Duncan Rouleau), (Duncan Rouleau, creator, Baymax), (Alan Tudyk, starring, Big Hero 6 (film)), (Steven T Seagle, creator, Baymax), (Big Hero 6 (film), series, Baymax)
Prefix-tuning	Baymax is a character in Big Hero 6 which stars Alan Tudyk. He was created by Steven T. Seagle and the American, Duncan Rouleau.
Fine-tuning	Alan Tudyk stars in the film Big Hero 6 in which Baymax is a character created by Steven T. Seagle and the American, Duncan Rouleau.
Reference	Baymax is a character who appeared in Big Hero 6 starring Alan Tudyk. It was created by Steven T Seagle and the American, Duncan Rouleau.
Source [Seen, City]	(Washington, D.C., capital, United States), (White Americans, ethnicGroup, United States), (United States, country, New Jersey), (New York City, largest City, United States), (New Jersey, isPartOf, Atlantic City)
Prefix-tuning	Washington D.C. is the capital of the United States where the largest city is New York City and the White Americans are an ethnic group. Atlantic City, New Jersey is also part of the United States.
Fine-tuning	Atlantic City, New Jersey is part of New Jersey in the United States. The capital city is Washington D.C. and one of the ethnic groups is White Americans.
Reference	New York City (NYC) is the largest U.S. city. Atlantic City, New Jersey are also part of the United States with its capital as Washington, DC and home to White Americans.

Table 6: Qualitative examples from WebNLG. The first 6 examples are from the unseen categories, labeled next to source; the last two examples are from the seen categories. For unseen categories, both prefix-tuning and fine-tuning tend to undergenerate (generated output do not cover full table contents) or generate untruthfully (generated output is inconsistent with table contents). In particular, prefix-tuning tends to undergenerate more often than generate untruthfully whereas fine-tuning tends to generate untruthfully. For seen categories, both perform fairly well in terms of coverage and truthfulness.